

Show that the Bellman-Ford algorithm, when run on the constraint graph for a system $Ax \leq b$ of difference constraints, minimizes the quantity $(\max \{x_i\} - \min \{x_i\})$ subject to $Ax \leq b$. Explain how this fact might come in handy if the algorithm is used to schedule construction jobs.

22.4-10

Suppose that every row in the matrix A of a linear program $Ax \leq b$ corresponds to a difference constraint, a single-variable constraint of the form $x_i \leq b_k$, or a single-variable constraint of the form $-x_i \leq b_k$. Show how to adapt the Bellman-Ford algorithm to solve this variety of constraint system.

22.4-11

Give an efficient algorithm to solve a system $Ax \leq b$ of difference constraints when all of the elements of b are real-valued and all of the unknowns x_i must be integers.

★ 22.4-12

Give an efficient algorithm to solve a system $Ax \leq b$ of difference constraints when all of the elements of b are real-valued and a specified subset of some, but not necessarily all, of the unknowns x_i must be integers.

22.5 Proofs of shortest-paths properties

Throughout this chapter, our correctness arguments have relied on the triangle inequality, upper-bound property, no-path property, convergence property, path-relaxation property, and predecessor-subgraph property. We stated these properties without proof on page 611. In this section, we prove them.

The triangle inequality

In studying breadth-first search (Section 20.2), we proved as Lemma 20.1 a simple property of shortest distances in unweighted graphs. The

triangle inequality generalizes the property to weighted graphs.

Lemma 22.10 (Triangle inequality)

Let $G = (V, E)$ be a weighted, directed graph with weight function $w : E \rightarrow \mathbb{R}$ and source vertex s . Then, for all edges $(u, v) \in E$,

$$\delta(s, v) \leq \delta(s, u) + w(u, v).$$

Proof Suppose that p is a shortest path from source s to vertex v . Then p has no more weight than any other path from s to v . Specifically, path p has no more weight than the particular path that takes a shortest path from source s to vertex u and then takes edge (u, v) .

Exercise 22.5-3 asks you to handle the case in which there is no shortest path from s to v . ■

Effects of relaxation on shortest-path estimates

The next group of lemmas describes how shortest-path estimates are affected by executing a sequence of relaxation steps on the edges of a weighted, directed graph that has been initialized by INITIALIZE-SINGLE-SOURCE.

Lemma 22.11 (Upper-bound property)

Let $G = (V, E)$ be a weighted, directed graph with weight function $w : E \rightarrow \mathbb{R}$. Let $s \in V$ be the source vertex, and let the graph be initialized by INITIALIZE-SINGLE-SOURCE(G, s). Then, $v.d \geq \delta(s, v)$ for all $v \in V$, and this invariant is maintained over any sequence of relaxation steps on the edges of G . Moreover, once $v.d$ achieves its lower bound $\delta(s, v)$, it never changes.

Proof We prove the invariant $v.d \geq \delta(s, v)$ for all vertices $v \in V$ by induction over the number of relaxation steps.

For the base case, $v.d \geq \delta(s, v)$ holds after initialization, since if $v.d = \infty$, then $v.d \geq \delta(s, v)$ for all $v \in V - \{s\}$, and since $s.d = 0 \geq \delta(s, s)$. (Note that $\delta(s, s) = -\infty$ if s is on a negative-weight cycle and that $\delta(s, s) = 0$ otherwise.)

For the inductive step, consider the relaxation of an edge (u, v) . By the inductive hypothesis, $x.d \geq \delta(s, x)$ for all $x \in V$ prior to the relaxation. The only d value that may change is $v.d$. If it changes, we have

$$\begin{aligned} v.d &= u.d + w(u, v) \\ &\geq \delta(s, u) + w(u, v) \text{ (by the inductive hypothesis)} \\ &\geq \delta(s, v) \text{ (by the triangle inequality),} \end{aligned}$$

and so the invariant is maintained.

The value of $v.d$ never changes once $v.d = \delta(s, v)$ because, having achieved its lower bound, $v.d$ cannot decrease since we have just shown that $v.d \geq \delta(s, v)$, and it cannot increase because relaxation steps do not increase d values. ■

Corollary 22.12 (No-path property)

Suppose that in a weighted, directed graph $G = (V, E)$ with weight function $w : E \rightarrow \mathbb{R}$, no path connects a source vertex $s \in V$ to a given vertex $v \in V$. Then, after the graph is initialized by INITIALIZE-SINGLE-SOURCE(G, s), we have $v.d = \delta(s, v) = \infty$, and this equation is maintained as an invariant over any sequence of relaxation steps on the edges of G .

Proof By the upper-bound property, we always have $\infty = \delta(s, v) \leq v.d$, and thus $v.d = \infty = \delta(s, v)$. ■

Lemma 22.13

Let $G = (V, E)$ be a weighted, directed graph with weight function $w : E \rightarrow \mathbb{R}$, and let $(u, v) \in E$. Then, immediately after edge (u, v) is relaxed by a call of RELAX(u, v, w), we have $v.d \leq u.d + w(u, v)$.

Proof If, just prior to relaxing edge (u, v) , we have $v.d > u.d + w(u, v)$, then $v.d = u.d + w(u, v)$ afterward. If, instead, $v.d \leq u.d + w(u, v)$ just before the relaxation, then neither $u.d$ nor $v.d$ changes, and so $v.d \leq u.d + w(u, v)$ afterward.

■

Lemma 22.14 (Convergence property)

Let $G = (V, E)$ be a weighted, directed graph with weight function $w : E \rightarrow \mathbb{R}$, let $s \in V$ be a source vertex, and let $s \rightsquigarrow u \rightarrow v$ be a shortest path in G for some vertices $u, v \in V$. Suppose that G is initialized by INITIALIZE-SINGLE-SOURCE(G, s) and then a sequence of relaxation steps that includes the call RELAX(u, v, w) is executed on the edges of G . If $u.d = \delta(s, u)$ at any time prior to the call, then $v.d = \delta(s, v)$ at all times after the call.

Proof By the upper-bound property, if $u.d = \delta(s, u)$ at some point prior to relaxing edge (u, v) , then this equation holds thereafter. In particular, after edge (u, v) is relaxed, we have

$$\begin{aligned} v.d &\leq u.d + w(u, v) && \text{(by Lemma 22.13)} \\ &= \delta(s, u) + w(u, v) \\ &= \delta(s, v) && \text{(by Lemma 22.1 on page 606).} \end{aligned}$$

The upper-bound property gives $v.d \geq \delta(s, v)$, from which we conclude that $v.d = \delta(s, v)$, and this equation is maintained thereafter.

■

Lemma 22.15 (Path-relaxation property)

Let $G = (V, E)$ be a weighted, directed graph with weight function $w : E \rightarrow \mathbb{R}$, and let $s \in V$ be a source vertex. Consider any shortest path $p = \langle v_0, v_1, \dots, v_k \rangle$ from $s = v_0$ to v_k . If G is initialized by INITIALIZE-SINGLE-SOURCE(G, s) and then a sequence of relaxation steps occurs that includes, in order, relaxing the edges $(v_0, v_1), (v_1, v_2), \dots, (v_{k-1}, v_k)$, then $v_k.d = \delta(s, v_k)$ after these relaxations and at all times afterward. This property holds no matter what other edge relaxations occur, including relaxations that are intermixed with relaxations of the edges of p .

Proof We show by induction that after the i th edge of path p is relaxed, we have $v_i.d = \delta(s, v_i)$. For the base case, $i = 0$, and before any edges of p

have been relaxed, we have from the initialization that $v_0.d = s.d = 0 = \delta(s, s)$. By the upper-bound property, the value of $s.d$ never changes after initialization.

For the inductive step, assume that $v_{i-1}.d = \delta(s, v_{i-1})$. What happens when edge (v_{i-1}, v_i) is relaxed? By the convergence property, after this relaxation, we have $v_i.d = \delta(s, v_i)$, and this equation is maintained at all times thereafter. ■

Relaxation and shortest-paths trees

We now show that once a sequence of relaxations has caused the shortest-path estimates to converge to shortest-path weights, the predecessor subgraph G_π induced by the resulting π values is a shortest-paths tree for G . We start with the following lemma, which shows that the predecessor subgraph always forms a rooted tree whose root is the source.

Lemma 22.16

Let $G = (V, E)$ be a weighted, directed graph with weight function $w : E \rightarrow \mathbb{R}$ let $s \in V$ be a source vertex, and assume that G contains no negative-weight cycles that are reachable from s . Then, after the graph is initialized by INITIALIZE-SINGLE-SOURCE(G, s), the predecessor subgraph G_π forms a rooted tree with root s , and any sequence of relaxation steps on edges of G maintains this property as an invariant.

Proof Initially, the only vertex in G_π is the source vertex, and the lemma is trivially true. Consider a predecessor subgraph G_π that arises after a sequence of relaxation steps. We first prove that G_π is acyclic. Suppose for the sake of contradiction that some relaxation step creates a cycle in the graph G_π . Let the cycle be $c = \langle v_0, v_1, \dots, v_k \rangle$, where $v_k = v_0$. Then, $v_i.\pi = v_{i-1}$ for $i = 1, 2, \dots, k$ and, without loss of generality, assume that relaxing edge (v_{k-1}, v_k) created the cycle in G_π .

We claim that all vertices on cycle c are reachable from the source vertex s . Why? Each vertex on c has a non-NIL predecessor, and so each vertex on c was assigned a finite shortest-path estimate when it was assigned its non-NIL π value. By the upper-bound property, each vertex on cycle c has a finite shortest-path weight, which means that it is reachable from s .

We'll examine the shortest-path estimates on cycle c immediately before the call $\text{RELAX}(v_{k-1}, v_k, w)$ and show that c is a negative-weight cycle, thereby contradicting the assumption that G contains no negative-weight cycles that are reachable from the source. Just before the call, we have $v_i.\pi = v_{i-1}$ for $i = 1, 2, \dots, k-1$. Thus, for $i = 1, 2, \dots, k-1$, the last update to $v_i.d$ was by the assignment $v_i.d = v_{i-1}.d + w(v_{i-1}, v_i)$. If $v_{i-1}.d$ changed since then, it decreased. Therefore, just before the call $\text{RELAX}(v_{k-1}, v_k, w)$, we have

$$v_i.d \geq v_{i-1}.d + w(v_{i-1}, v_i) \quad \text{for all } i = 1, 2, \dots, k-1. \quad (22.11)$$

Because $v_k.\pi$ is changed by the call $\text{RELAX}(v_{k-1}, v_k, w)$, immediately beforehand we also have the strict inequality

$$v_k.d > v_{k-1}.d + w(v_{k-1}, v_k):$$

Summing this strict inequality with the $k-1$ inequalities (22.11), we obtain the sum of the shortest-path estimates around cycle c :

$$\begin{aligned} \sum_{i=1}^k v_i.d &> \sum_{i=1}^k (v_{i-1}.d + w(v_{i-1}, v_i)) \\ &= \sum_{i=1}^k v_{i-1}.d + \sum_{i=1}^k w(v_{i-1}, v_i). \end{aligned}$$

But

$$\sum_{i=1}^k v_i.d = \sum_{i=1}^k v_{i-1}.d,$$

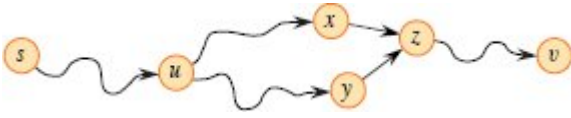


Figure 22.9 Showing that a simple path in G_π from source vertex s to vertex v is unique. If G_π contains two paths p_1 ($s \rightsquigarrow u \rightsquigarrow x \rightarrow z \rightsquigarrow v$) and p_2 ($s \rightsquigarrow u \rightsquigarrow y \rightarrow z \rightsquigarrow v$), where $x \neq y$, then $z.\pi = x$ and $z.\pi = y$, a contradiction.

since each vertex in the cycle c appears exactly once in each summation. This equation implies

$$0 > \sum_{i=1}^k w(v_{i-1}, v_i) .$$

Thus, the sum of weights around the cycle c is negative, which provides the desired contradiction.

We have now proven that G_π is a directed, acyclic graph. To show that it forms a rooted tree with root s , it suffices (see Exercise B.5-2 on page 1175) to prove that for each vertex $v \in V_\pi$, there is a unique simple path from s to v in G_π .

The vertices in V_π are those with non-NIL values, plus s . Exercise 22.5-6 asks you to prove that a path from s exists to each vertex in V_π .

To complete the proof of the lemma, we now show that for any vertex $v \in V_\pi$, the graph G_π contains at most one simple path from s to v . Suppose otherwise. That is, suppose that, as Figure 22.9 illustrates, G_π contains two simple paths from s to some vertex v : p_1 , which we decompose into $s \rightsquigarrow u \rightsquigarrow x \rightarrow z \rightsquigarrow v$, and p_2 , which we decompose into $s \rightsquigarrow u \rightsquigarrow y \rightarrow z \rightsquigarrow v$, where $x \neq y$ (though u could be s and z could be v). But then, $z.\pi = x$ and $z.\pi = y$, which implies the contradiction that $x = y$. We conclude that G_π contains a unique simple path from s to v , and thus G_π forms a rooted tree with root s . ■

We can now show that if all vertices have been assigned their true shortest-path weights after a sequence of relaxation steps, then the

predecessor subgraph G_π is a shortest-paths tree.

Lemma 22.17 (Predecessor-subgraph property)

Let $G = (V, E)$ be a weighted, directed graph with weight function $w : E \rightarrow \mathbb{R}$, let $s \in V$ be a source vertex, and assume that G contains no negative-weight cycles that are reachable from s . Then, after a call to INITIALIZE-SINGLE-SOURCE(G, s) followed by any sequence of relaxation steps on edges of G that produces $v.d = \delta(s, v)$ for all $v \in V$, the predecessor subgraph G_π is a shortest-paths tree rooted at s .

Proof We must prove that the three properties of shortest-paths trees given on page 608 hold for G_π . To show the first property, we must show that V_π is the set of vertices reachable from s . By definition, a shortest-path weight $\delta(s, v)$ is finite if and only if v is reachable from s , and thus the vertices that are reachable from s are exactly those with finite d values. But a vertex $v \in V - \{s\}$ has been assigned a finite value for $v.d$ if and only if $v.\pi \neq \text{NIL}$, since both assignments occur in RELAX. Thus, the vertices in V_π are exactly those reachable from s .

The second property, that G_π forms a rooted tree with root s , follows directly from Lemma 22.16.

It remains, therefore, to prove the last property of shortest-paths trees: for each vertex $v \in V_\pi$, the unique simple path $s \stackrel{p}{\rightsquigarrow} v$ in G_π is a shortest path from s to v in G . Let $p = \langle v_0, v_1, \dots, v_k \rangle$, where $v_0 = s$ and $v_k = v$. Consider an edge (v_{i-1}, v_i) in path p . Because this edge belongs to G_π , the last relaxation that changed $v_i.d$ must have been of this edge. After that relaxation, we had $v_i.d = v_{i-1}.d + (v_{i-1}, v_i)$. Subsequently, an edge entering v_{i-1} could have been relaxed, causing $v_{i-1}.d$ to decrease further, but without changing $v_i.d$. Therefore, we have $v_i.d \geq v_{i-1}.d + w(v_{i-1}, v_i)$. Thus, for $i = 1, 2, \dots, k$, we have both $v_i.d = \delta(s, v_i)$ and $v_i.d \geq v_{i-1}.d + w(v_{i-1}, v_i)$, which together imply $w(v_{i-1}, v_i) \leq \delta(s, v_i) - \delta(s, v_{i-1})$. Summing the weights along path p yields

$$\begin{aligned}
w(p) &= \sum_{i=1}^k w(v_{i-1}, v_i) \\
&\leq \sum_{i=1}^k (\delta(s, v_i) - \delta(s, v_{i-1})) \\
&= \delta(s, v_k) - \delta(s, v_0) \quad (\text{because the sum telescopes}) \\
&= \delta(s, v_k) \quad (\text{because } \delta(s, v_0) = \delta(s, s) = 0) .
\end{aligned}$$

Thus, we have $w(p) \leq \delta(s, v_k)$. Since $\delta(s, v_k)$ is a lower bound on the weight of any path from s to v_k , we conclude that $w(p) = \delta(s, v_k)$, and p is a shortest path from s to $v = v_k$. ■

Exercises

22.5-1

Give two shortest-paths trees for the directed graph of Figure 22.2 on page 609 other than the two shown.

22.5-2

Give an example of a weighted, directed graph $G = (V, E)$ with weight function $w : E \rightarrow \mathbb{R}$ and source vertex s such that G satisfies the following property: For every edge $(u, v) \in E$, there is a shortest-paths tree rooted at s that contains (u, v) and another shortest-paths tree rooted at s that does not contain (u, v) .

22.5-3

Modify the proof of Lemma 22.10 to handle cases in which shortest-path weights are ∞ or $-\infty$.

22.5-4

Let $G = (V, E)$ be a weighted, directed graph with source vertex s , and let G be initialized by INITIALIZE-SINGLE-SOURCE(G, s). Prove that if a sequence of relaxation steps sets $s.\pi$ to a non-NIL value, then G contains a negative-weight cycle.

22.5-5

Let $G = (V, E)$ be a weighted, directed graph with no negative-weight edges. Let $s \in V$ be the source vertex, and suppose that $v.\pi$ is allowed to be the predecessor of v on *any* shortest path to v from source s if $v \in V - \{s\}$ is reachable from s , and NIL otherwise. Give an example of such a graph G and an assignment of π values that produces a cycle in G_π . (By Lemma 22.16, such an assignment cannot be produced by a sequence of relaxation steps.)

22.5-6

Let $G = (V, E)$ be a weighted, directed graph with weight function $w : E \rightarrow \mathbb{R}$ and no negative-weight cycles. Let $s \in V$ be the source vertex, and let G be initialized by INITIALIZE-SINGLE-SOURCE(G, s). Use induction to prove that for every vertex $v \in V_\pi$, there exists a path from s to v in G_π and that this property is maintained as an invariant over any sequence of relaxations.

22.5-7

Let $G = (V, E)$ be a weighted, directed graph that contains no negative-weight cycles. Let $s \in V$ be the source vertex, and let G be initialized by INITIALIZE-SINGLE-SOURCE(G, s). Prove that there exists a sequence of $|V| - 1$ relaxation steps that produces $v.d = \delta(s, v)$ for all $v \in V$.

22.5-8

Let G be an arbitrary weighted, directed graph with a negative-weight cycle reachable from the source vertex s . Show how to construct an infinite sequence of relaxations of the edges of G such that every relaxation causes a shortest-path estimate to change.

Problems

22-1 *Yen's improvement to Bellman-Ford*

The Bellman-Ford algorithm does not specify the order in which to relax edges in each pass. Consider the following method for deciding upon the order. Before the first pass, assign an arbitrary linear order $v_1,$